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Comprehensive Question Bank: Unit 7 — Applications of AI

Section A: Multiple-Choice Questions (MCQs)

Topic 7.1: Introduction to Artificial Intelligence (AI)

1. What does AI stand for?
A) Automatic Intelligence
B) Artificial Intelligence
C) Active Interface
D) Applied Information
2. In 1950, which scientist proposed the question "Can machines think?" and designed a test to evaluate it?
A) Alan Turing
B) Joseph Weizenbaum
C) Bill Gates
D) Isaac Newton
3. What is the name of the test in which a human judge tries to tell a human and a machine apart through conversation?
A) The Machine Test
B) The Turing Test
C) The Weizenbaum Test
D) The Logic Test
4. Which of the following best defines Artificial Intelligence?
A) A machine that only performs arithmetic
B) A device used only for gaming
C) Computer software that learns from data to perform tasks that normally require human intelligence
D) A type of hardware component inside a computer
5. What makes AI different from a normal calculator program?
A) AI is always slower
B) AI never makes mistakes
C) AI learns patterns from data instead of following one fixed rule
D) AI cannot be used on phones

6. Which of the following is NOT listed as a common area where AI is used?
- A) Mobile apps
 - B) Online shopping
 - C) Health care
 - D) Manual paper filing**
7. When you talk to a voice assistant or use a website chatbot, what are you actually using?
- A) A basic calculator
 - B) Artificial Intelligence**
 - C) A spreadsheet
 - D) A word processor
8. According to the chapter, AI can now do which of the following creative tasks?
- A) Only solve math problems
 - B) Only play music
 - C) Write stories, solve math problems, and create art**
 - D) Only take photographs
9. What is the key idea that AI helps computers do without being told every single step?
- A) Print documents
 - B) Make smart decisions on their own**
 - C) Charge their battery
 - D) Connect to Wi-Fi
10. Which of these is an example of AI helping in transportation, as mentioned in the chapter?
- A) A car radio playing music
 - B) A maps app predicting the fastest route**
 - C) A car's manual gear shift
 - D) A fuel gauge showing petrol level
-

Topic 7.1.1: Impact of AI on Jobs and Work

1. What job became largely obsolete due to automatic telephone switching systems?
- A) Data scientist
 - B) Telephone switchboard operator**
 - C) AI developer
 - D) Machine learning engineer
2. According to the chapter, what is one major effect of AI in factories?
- A) Factories have stopped using machines completely
 - B) Robots and machines now do jobs once done by human experts**
 - C) All human workers have been replaced by AI in every field
 - D) Factories now require more paperwork

3. Which of the following is a new job role that has grown because of AI?
 - A) Switchboard operator
 - B) Machine Learning Engineer**
 - C) Typewriter repair technician
 - D) Telegram operator
 4. What is a "Data Scientist," according to the chapter's context?
 - A) A person who repairs computer hardware
 - B) A person who studies data to find patterns**
 - C) A person who only cleans office spaces
 - D) A person who sells computers
 5. Which statement best reflects the chapter's view on AI's overall impact on jobs?
 - A) AI destroys jobs and creates nothing in return
 - B) AI often replaces repetitive tasks while creating new, more technical roles**
 - C) AI has no impact on employment at all
 - D) AI can only be used by large factories, not offices
 6. Machines used in factories because of AI can typically do which of the following, compared to humans?
 - A) Work slower and make more mistakes
 - B) Work faster, longer, and with fewer mistakes**
 - C) Work only during the day
 - D) Work only one hour per day
-

Topic 7.2: Data and AI

1. What is "data" in the context of Artificial Intelligence?
 - A) Only images stored on a phone
 - B) Information — numbers, words, images, sounds — fed into a computer**
 - C) A type of hardware chip
 - D) A programming language
2. Data is described in the chapter as being as important to AI as what is to a student?
 - A) A calculator
 - B) Books they read to learn**
 - C) A school uniform
 - D) A bus ride to school

3. What can happen if an AI system is trained with poor-quality data?
 - A) It always works perfectly
 - B) It becomes faster automatically
 - C) It can give wrong answers or make unfair decisions**
 - D) It stops needing electricity
 4. What does an AI system look for within large amounts of data to improve its decisions?
 - A) File sizes
 - B) Patterns**
 - C) Colors only
 - D) Battery levels
-

Topic 7.2.1: Automatic Data Collection

1. What is "automatic data collection"?
 - A) Typing information into a computer by hand
 - B) Gathering information without a person entering it manually**
 - C) Deleting old files automatically
 - D) Printing documents automatically
2. Which of the following is an example of automatic data collection mentioned in the chapter?
 - A) A teacher writing on a whiteboard
 - B) A mobile phone collecting location, usage, and search history**
 - C) A student handwriting an essay
 - D) A librarian stamping a book
3. What is one major benefit of automatic data collection for AI systems?
 - A) It makes systems require constant human typing
 - B) It allows systems to adjust and respond to user needs quickly**
 - C) It increases the number of errors
 - D) It slows down every device
4. Which of these devices was used as an example of automatic data collection in the chapter?
 - A) A wall clock
 - B) A smart wristwatch/fitness watch**
 - C) A paper notebook
 - D) A whiteboard marker
5. What type of information do online shopping websites typically collect automatically?
 - A) Only the user's name
 - B) User clicks, views, and browsing behavior**
 - C) Only the user's home address
 - D) Nothing at all

Topic 7.2.2: Types of Data — Quality and Quantity

1. What does "quantity" refer to in the context of AI training data?
 - A) How correct the data is
 - B) How much data is available**
 - C) How colorful the data is
 - D) How the data is stored
 2. What does "quality" refer to in the context of AI training data?
 - A) The size of the data file
 - B) How useful and correct the data is**
 - C) The number of devices used to collect the data
 - D) The speed at which data is collected
 3. In the chapter's dog-training analogy, why does the dog fail to recognize a brown cat?
 - A) The dog is not smart enough
 - B) It was trained using only pictures of white cats, so its training data was incomplete**
 - C) The dog was too old to learn
 - D) Cats cannot be recognized by any animal
 4. According to the chapter, what happens if data has mistakes, missing values, or confusing labels?
 - A) The AI ignores the data automatically
 - B) The AI may learn the wrong things**
 - C) The AI becomes faster
 - D) The AI shuts down permanently
 5. What combination does AI need to learn most effectively, according to the chapter?
 - A) Small and inaccurate data
 - B) Big (quantity) and accurate (quality) data**
 - C) Only large quantities of any kind of data
 - D) Only a few examples, regardless of accuracy
 6. A dataset of 10,000 photos where 9,500 are repeated images of the same three cats is an example of:
 - A) High quality and high quantity data
 - B) Large quantity but poor quality/variety of data**
 - C) Small but perfectly balanced data
 - D) Data with no images at all
-

Topic 7.3.1: Natural Language Processing (NLP)

1. What does NLP stand for?
 - A) Natural Learning Program
 - B) Natural Language Processing**
 - C) Network Language Protocol
 - D) National Language Policy
2. Who created ELIZA, one of the first chatbots, in 1966?
 - A) Alan Turing
 - B) Joseph Weizenbaum**
 - C) Bill Gates
 - D) Amazon engineers
3. What did ELIZA simulate?
 - A) A weather forecaster
 - B) A psychotherapist**
 - C) A robot arm
 - D) A self-driving car
4. How did ELIZA generate its responses?
 - A) By understanding emotions deeply
 - B) By using simple keyword pattern-matching**
 - C) By using cameras to see the user
 - D) By connecting to the internet
5. Which of the following is an example of NLP mentioned in the chapter?
 - A) An online calculator
 - B) Voice typing**
 - C) File copying
 - D) Image editing
6. In the NLP pipeline walkthrough, what is the first step when processing the sentence "I love this camera!"?
 - A) Sentiment detection
 - B) Tokenization — breaking the sentence into individual pieces**
 - C) Output generation
 - D) GPS tracking
7. What is "tokenization" in NLP?
 - A) Converting text into video
 - B) Breaking a sentence into smaller individual pieces (tokens)**
 - C) Deleting a sentence completely
 - D) Translating a sentence into another language

8. In the example sentence "I love this camera!", what sentiment would the AI most likely detect?
- A) Negative sentiment
 - B) Positive sentiment**
 - C) Neutral sentiment
 - D) No sentiment detected
9. Which of the following best describes how NLP helps in customer service?
- A) It replaces all human customer service permanently
 - B) It helps answer common questions without needing human help**
 - C) It only works with images
 - D) It cannot understand written text
10. What kind of technology helps a search engine understand a typed question and reply correctly?
- A) Robotics
 - B) NLP**
 - C) GPS
 - D) Radar
-

Topic 7.3.2: Robotics in Everyday Life

1. What was Unimate, deployed by General Motors in 1961?
- A) The first chatbot
 - B) The first industrial robot**
 - C) The first voice assistant
 - D) The first self-driving car
2. What task did Unimate perform on the GM assembly line?
- A) Painting cars only
 - B) Lifting and stacking hot die-castings**
 - C) Selling cars to customers
 - D) Driving cars off the factory line
3. What is a "robot," as defined in the chapter?
- A) A software application only
 - B) A machine designed to perform tasks like humans**
 - C) A type of data file
 - D) A voice assistant only

4. According to the chapter, why are robots especially useful for certain tasks?
- A) They require constant supervision
 - B) They get tired slower than humans but still need breaks
 - C) They do not get tired and can work with great accuracy**
 - D) They can only work indoors
5. Which of the following is mentioned as a real-world use of robots?
- A) Writing novels
 - B) Helping in surgeries**
 - C) Teaching history classes
 - D) Composing music
6. According to the "Did You Know?" note, where are robots now used to serve food and clean tables?
- A) Hospitals only
 - B) Some restaurants**
 - C) Airports only
 - D) Universities only
7. In the robot task walkthrough, what is the first step in analyzing a robot's behavior?
- A) Predicting the failure case
 - B) Watching/observing the task the robot performs**
 - C) Writing a summary
 - D) Identifying ethical issues
-

Topic 7.3.3: Speech Technology and Voice Control

1. What does speech technology primarily allow computers to do?
- A) Only display text on a screen
 - B) Listen to and understand spoken words**
 - C) Only take photographs
 - D) Only browse the internet
2. According to the chapter, what can smartphones now do using speech technology, without touching the screen?
- A) Charge the battery
 - B) Unlock using your voice**
 - C) Install new apps
 - D) Change the wallpaper

3. Which of the following is an example of speech technology used in smart homes?
- A) Typing a report
 - B) Controlling lights or devices by speaking**
 - C) Printing a document
 - D) Charging a laptop
4. In healthcare, how can speech technology assist doctors, according to the chapter?
- A) By performing surgery independently
 - B) By helping doctors record notes without typing**
 - C) By replacing all doctors
 - D) By diagnosing diseases without any data
5. What does the "Speak and See" activity ask students to test?
- A) How fast a phone can charge
 - B) How accurately a voice-to-text tool converts speech into text**
 - C) How to install new apps
 - D) How to take better photographs
6. What can cars respond to, according to the chapter's example of voice control?
- A) Only steering wheel movement
 - B) Voice commands to turn on music, make calls, or give directions**
 - C) Only headlights
 - D) Only seatbelt sensors
-

Topic 7.3.4: Recommendation Systems

1. What method did Amazon introduce in 1998 to predict what shoppers might want next?
- A) Manual Product Sorting
 - B) Item-to-Item Collaborative Filtering**
 - C) Direct Advertisement Placement
 - D) Random Suggestion Generator
2. What is a "recommendation system"?
- A) A system that only stores files
 - B) A part of AI that gives suggestions based on users' likes and past actions**
 - C) A system that deletes browsing history
 - D) A tool used only for typing documents
3. In the recommendation pipeline walkthrough, what is the first step after a user watches a cooking video?
- A) Filtering

- B) Logging the user action**
C) Showing the top 3 predictions
D) Deleting the video
4. What does "Pattern Matching" refer to in a recommendation system pipeline?
A) Comparing font styles used on the screen
B) Comparing a user's action to millions of other users with similar behavior
C) Matching passwords for login
D) Matching screen resolutions
5. According to the chapter, what does YouTube's "Videos you may like" feature demonstrate?
A) Robotics in action
B) AI learning your interests through a recommendation system
C) A speech recognition tool
D) An autonomous vehicle sensor
6. What must recommendation systems be careful about, according to the chapter?
A) Increasing screen brightness
B) Protecting privacy and avoiding repeatedly showing the same content
C) Reducing internet speed
D) Removing all suggestions completely
7. If someone views school bags on a shopping website, what might a recommendation system suggest next?
A) Unrelated electronics only
B) Other bags or related items
C) Random items with no connection
D) Nothing at all
-

Topic 7.3.5: Autonomous Vehicles (Self-Driving Vehicles)

1. What is an "autonomous vehicle"?
A) A vehicle that requires two drivers
B) A vehicle that can move on the road without a human driver
C) A vehicle that only works on highways
D) A vehicle powered only by solar energy
2. Which of the following sensors is used by self-driving cars to detect the exact location on a map?
A) Radar
B) Camera
C) GPS (Global Positioning System)
D) LiDAR

3. Which sensor is primarily used to detect the distance and speed of nearby vehicles?
- A) Camera
 - B) Radar**
 - C) GPS
 - D) Microphone
4. What do cameras in a self-driving car primarily detect?
- A) Only the car's internal temperature
 - B) Traffic lights, road signs, and lane markings**
 - C) Only tire pressure
 - D) Only fuel levels
5. What is one major benefit of autonomous vehicles mentioned in the chapter?
- A) They eliminate the need for roads
 - B) They can reduce accidents caused by human error like distraction or tiredness**
 - C) They require no testing at all
 - D) They can only be used by professional racers
6. Which group of people is mentioned as potentially benefiting most from autonomous vehicles' independence?
- A) Professional drivers only
 - B) The elderly or disabled who are unable to drive**
 - C) Vehicle manufacturers only
 - D) Traffic police only
7. Which of the following is listed as a challenge of autonomous vehicles?
- A) They are too cheap to produce
 - B) Concerns about who is responsible if a self-driving car causes an accident**
 - C) They cannot be tested at all
 - D) They require no sensors
8. Who must work together to improve safety and create fair rules for autonomous vehicles, according to the chapter?
- A) Only car owners
 - B) Engineers, governments, and companies**
 - C) Only pedestrians
 - D) Only vehicle manufacturers
-

Topic 7.4: Benefits and Challenges of AI Applications

1. According to the chapter, how does AI help in offices?
A) By replacing all office furniture
B) By managing files and answering questions
C) By removing the need for electricity
D) By eliminating all office jobs immediately
 2. How can AI help students in education, according to the chapter?
A) By writing all their exams for them
B) Through smart learning apps that adjust to their level
C) By replacing every teacher immediately
D) By deleting their study materials
 3. Which of the following fields is mentioned as benefiting from AI's efficiency improvements?
A) Ancient history research only
B) Agriculture, banking, and transport
C) Handwriting calligraphy only
D) Paper manufacturing only
-

Topic 7.4.1: Limitations and Challenges of AI Applications

1. What can cause an AI system, like a chatbot, to misunderstand a question?
A) Having too much electricity
B) Unclear or ambiguous language in the question
C) Being connected to Wi-Fi
D) Being too colorful
2. In which fields does the chapter warn that small AI mistakes can cause big problems?
A) Gaming and entertainment
B) Law and health
C) Music and art
D) Sports only
3. What is one challenge related to jobs mentioned as a limitation of AI?
A) AI always increases the number of jobs
B) AI can replace human workers in fields like customer service, data entry, and clerical work
C) AI only works with government jobs
D) AI has no relationship with employment at all

4. According to the "Did You Know?" note in this section, what can cause AI systems to make mistakes?
- A) Being used too often
 - B) Being trained on incomplete or biased data**
 - C) Being connected to too many devices
 - D) Being too new
-

Topic 7.5: Ethics, Fairness, and Social Impact of AI

1. What does "ethics" in AI mean, according to the chapter?
 - A) Making AI systems run faster
 - B) Using AI in a way that is fair, honest, and safe for everyone**
 - C) Making AI systems more expensive
 - D) Making AI systems work only for developers
 2. What happens if an AI system gives better service to one group of people and ignores others?
 - A) This is considered acceptable and efficient
 - B) This is considered unfair**
 - C) This has no ethical implications
 - D) This is required by all AI systems
 3. What three qualities should AI be developed with, according to the chapter?
 - A) Speed, cost, and popularity
 - B) Fairness, safety, and respect for human rights**
 - C) Complexity, size, and color
 - D) Price, weight, and brand name
-

Topic 7.5.1: Bias in AI and Algorithms

1. What is "bias" in AI, as defined in the chapter?
 - A) A random computer error with no cause
 - B) Unfair results caused by the data a system was trained on**
 - C) A feature that makes AI faster
 - D) A type of software license
2. In the chapter's real-world example, why did some automatic soap dispensers fail to detect dark skin tones?
 - A) The dispensers were broken from the factory
 - B) Their infrared sensors were calibrated and tested using mostly light-skinned hands**
 - C) The dispensers did not have any sensors at all
 - D) The soap itself was defective

3. Why might a face recognition tool not work well for all skin colors?
 - A) Because cameras cannot see color at all
 - B) Because it was trained mostly on images of only one skin type**
 - C) Because face recognition does not use data
 - D) Because skin color affects internet speed
 4. What is described as a big step toward making AI more fair and helpful for everyone?
 - A) Increasing the price of AI systems
 - B) Removing bias from training data**
 - C) Reducing the number of users
 - D) Making AI systems slower
 5. In the mini bias audit walkthrough, what is the first step recommended?
 - A) Proposing a fix immediately
 - B) Stating the AI's purpose**
 - C) Ignoring the training data
 - D) Deleting the system
 6. If a facial recognition security system is trained on data containing 90% male photos, what is the likely outcome?
 - A) It will work perfectly for everyone
 - B) It may fail to correctly recognize female employees**
 - C) It will stop working for everyone
 - D) It will automatically fix itself over time
-

Topic 7.5.2: Fairness and Transparency in AI

1. What does "fairness" in AI mean, according to the chapter?
 - A) Making AI systems expensive for everyone
 - B) Treating all users equally, no matter their race, gender, age, or background**
 - C) Giving priority to one specific group of users
 - D) Making AI systems run only on expensive devices
2. What does "transparency" in AI mean?
 - A) Making an AI system invisible to users
 - B) People understanding how an AI system works and why it made a certain decision**
 - C) Making an AI system completely free of charge
 - D) Hiding all information about how decisions are made

3. According to the chapter, if an AI rejects a loan application, what right do users have?
- A) No rights at all
 - B) The right to know what rules or steps the system followed**
 - C) The right to reject the AI system entirely
 - D) The right to change the AI's code themselves
4. In which of the following areas is fairness in AI decision-making especially important, according to the chapter?
- A) Video game high scores
 - B) School admissions, job interviews, and legal cases**
 - C) Weather forecasting only
 - D) Music recommendations only
5. What does transparency help avoid, according to the chapter?
- A) Faster processing speeds
 - B) Mistakes, while building confidence that a system is working honestly**
 - C) The need for any data at all
 - D) The use of sensors in AI systems
-

Topic 7.5.3: Privacy, Data Security, and Surveillance

1. What does "privacy" mean in the context of AI, according to the chapter?
- A) Sharing all personal information publicly
 - B) Keeping personal information safe and not sharing it without permission**
 - C) Deleting all data immediately
 - D) Making all data available to everyone
2. What does "data security" mean?
- A) Making data available to hackers
 - B) Protecting information from hackers, viruses, or other attacks**
 - C) Deleting all stored information
 - D) Sharing passwords publicly
3. What is "surveillance," as described in the chapter?
- A) A method of storing files
 - B) When systems or cameras watch people's actions**
 - C) A type of chatbot
 - D) A method of translating languages

4. What can happen if data is not protected properly, according to the chapter?
 - A) It automatically becomes more accurate
 - B) It can be stolen or misused**
 - C) It disappears completely
 - D) It becomes public property legally
 5. According to the chapter, what can happen if surveillance is used without reason, all the time?
 - A) It always improves safety with no downsides
 - B) It can harm freedom and privacy**
 - C) It has no effect on people at all
 - D) It automatically becomes illegal everywhere
-

Topic 7.5.4: AI and Social Equity (Fairness in Society)

1. According to the chapter, which groups should AI help equally?
 - A) Only wealthy and educated groups
 - B) All people, including the poor, disabled, and those in remote areas**
 - C) Only people living in cities
 - D) Only large businesses
 2. What problem can occur if new AI technologies are only available to rich or educated groups?
 - A) No problem occurs at all
 - B) It creates a gap between people who benefit and those who are left behind**
 - C) It automatically becomes free for everyone
 - D) It eliminates the need for social equity discussions
 3. What should AI tools be designed to ensure, according to the chapter, for social equity?
 - A) Only a few people benefit
 - B) Everyone can benefit, not just a few**
 - C) Only city residents benefit
 - D) Only technology companies benefit
-

Topic 7.6: Responsibility of AI Designers

1. According to the chapter, who creates the rules that AI systems follow?
 - A) The AI systems create their own rules independently
 - B) Human designers**
 - C) Government officials exclusively
 - D) No one; rules appear automatically

2. If an AI system makes a wrong decision or treats someone unfairly, where does the mistake usually originate, according to the chapter?
 - A) It is always a random, unexplainable glitch
 - B) The way the system was designed**
 - C) The user's device settings
 - D) The weather conditions
 3. According to the chapter, building trustworthy AI is described as being more than just a technical job — it is also a:
 - A) Marketing strategy
 - B) Moral duty**
 - C) Legal requirement only in some countries
 - D) Hobby for engineers
-

Topic 7.6.1: Following Ethics in AI

1. What must AI designers make sure their systems do NOT do, according to the chapter?
 - A) Process any data at all
 - B) Hurt people, spread false information, or treat anyone unfairly**
 - C) Operate on mobile devices
 - D) Use any form of automation
 2. What does "being honest about how the AI system works" refer to, according to the chapter?
 - A) Hiding all information from users
 - B) Ensuring users know what data is used and how decisions are made**
 - C) Making the system's code a permanent secret
 - D) Avoiding all forms of user communication
 3. What does ethical design help build, according to the chapter?
 - A) Faster internet connections
 - B) Trust between people and technology**
 - C) Cheaper hardware components
 - D) Larger data storage capacity
-

Section B: Short Answer Questions (SQs)

Topic 7.1: Introduction to Artificial Intelligence (AI)

1. What is Artificial Intelligence (AI)?
2. Who proposed the question "Can machines think?" and in what year?

3. Explain briefly what the Turing Test evaluates.
4. Differentiate between how a traditional calculator program works and how an AI system works.
5. List three areas of daily life where AI is used, according to the chapter.
6. Explain briefly why using a voice assistant counts as "using AI."

Topic 7.1.1: Impact of AI on Jobs and Work

1. Explain briefly how AI has changed factory work.
2. Differentiate between the role of a switchboard operator (before automation) and a data scientist (after AI).
3. List two new job roles that have grown because of AI.
4. Predict one job that might be affected by AI in the next decade, and explain why.

Topic 7.2: Data and AI

1. Define "data" in the context of Artificial Intelligence.
2. Explain briefly why data is considered the most important part of AI.
3. State what can happen if an AI system is trained using poor-quality data.
4. Explain briefly what a "pattern" means in the context of AI learning from data.

Topic 7.2.1: Automatic Data Collection

1. Define "automatic data collection."
2. List two examples of devices or apps that collect data automatically.
3. Explain briefly one benefit of automatic data collection for AI systems.
4. Identify one type of data a fitness watch might collect automatically.

Topic 7.2.2: Types of Data — Quality and Quantity

1. Differentiate between "quantity" and "quality" of data in AI.
2. Explain briefly, using the dog-and-cat analogy, how incomplete training data can cause errors.
3. Predict the outcome of training an AI system using data with many missing or incorrect labels.
4. State why AI needs both quantity and quality of data to perform well.

Topic 7.3.1: Natural Language Processing (NLP)

1. Define Natural Language Processing (NLP).
2. Who created ELIZA, and what did it simulate?
3. Explain briefly how ELIZA generated its responses without truly "understanding" language.
4. Map out the four steps of the NLP pipeline used to process the sentence "I love this camera!"
5. Differentiate between "tokenization" and "sentiment detection" in NLP.

6. Explain briefly how NLP helps in customer service.

Topic 7.3.2: Robotics in Everyday Life

1. Define "robotics" as used in the chapter.
2. What was Unimate, and what task did it perform in 1961?
3. Explain briefly why robots are useful for repeated or dangerous tasks.
4. List two real-world examples of robots mentioned in the chapter.
5. Identify one sensor a factory robot might use, and explain what it detects.

Topic 7.3.3: Speech Technology and Voice Control

1. Define "speech technology."
2. Explain briefly how speech technology is used in smart homes.
3. State one way speech technology assists doctors in healthcare.
4. Differentiate between how speech technology is used in cars versus in customer service.

Topic 7.3.4: Recommendation Systems

1. Define "recommendation system."
2. Explain briefly what Amazon's "Item-to-Item Collaborative Filtering," introduced in 1998, does.
3. Map out the four steps of a recommendation system pipeline, from user action to top 3 predictions.
4. State one risk or concern associated with recommendation systems.

Topic 7.3.5: Autonomous Vehicles (Self-Driving Vehicles)

1. Define "autonomous vehicle."
2. List the four types of sensors used by self-driving cars and briefly state what each one detects.
3. Explain briefly one benefit of autonomous vehicles.
4. Explain briefly one challenge or concern related to autonomous vehicles.
5. Predict what sensor data a self-driving car should prioritize in poor weather conditions.

Topic 7.4: Benefits and Challenges of AI Applications

1. Explain briefly how AI helps in offices.
2. State one way AI assists students in education.
3. List two fields, other than education, that benefit from AI efficiency, according to the chapter.

Topic 7.4.1: Limitations and Challenges of AI Applications

1. Explain briefly why a chatbot might misunderstand a question.

2. State why small AI mistakes in law or health can be especially dangerous.
3. Identify one job-related challenge caused by AI, according to the chapter.

Topic 7.5: Ethics, Fairness, and Social Impact of AI

1. Define "ethics" in the context of AI.
2. Explain briefly why it is unfair for an AI system to give better service to one group of people.
3. List three qualities AI should be developed with, according to the chapter.

Topic 7.5.1: Bias in AI and Algorithms

1. Define "bias" in AI.
2. Explain briefly the real-world example of automatic soap dispensers and bias.
3. State why a face recognition tool might not work well for all skin tones.
4. Identify the first step in conducting a mini bias audit, as outlined in the chapter.

Topic 7.5.2: Fairness and Transparency in AI

1. Define "fairness" in AI.
2. Define "transparency" in AI.
3. Differentiate between fairness and transparency using an example.
4. Explain briefly why transparency is important when an AI rejects a loan application.

Topic 7.5.3: Privacy, Data Security, and Surveillance

1. Define "privacy" in the context of AI.
2. Define "data security."
3. Differentiate between "surveillance" and "data security."
4. Explain briefly what can happen if surveillance is used without reason, all the time.

Topic 7.5.4: AI and Social Equity (Fairness in Society)

1. Explain briefly what "AI and Social Equity" means.
2. State one problem that can occur if AI technologies are only available to wealthy groups.
3. Identify what AI tools should be designed to ensure, according to the chapter.

Topic 7.6: Responsibility of AI Designers

1. Explain briefly why AI designers carry a big responsibility.
2. State where the root cause of an unfair AI decision usually comes from, according to the chapter.
3. Explain briefly why the chapter describes building trustworthy AI as a "moral duty."

Topic 7.6.1: Following Ethics in AI

1. Explain briefly what "following ethics in AI" means.
 2. List two things AI designers must ensure their systems do not do.
 3. State what ethical design helps build between people and technology.
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Section C: Long / Extensive Questions (LQs)

Topic 7.1 & 7.1.1: Introduction to AI and Its Impact on Jobs

1. Discuss in detail the concept of Artificial Intelligence and its role in modern computer science.
 - a) Explain the Turing Test and why Alan Turing's question "Can machines think?" was historically significant.
 - b) Define AI in plain language and explain how it differs from traditional, rule-based computer programs.
 - c) List and explain at least four real-world areas where AI is applied in daily life.
 2. Analyze the impact of Artificial Intelligence on jobs and the workforce.
 - a) Explain, using the telephone switchboard operator example, how automation historically changed employment.
 - b) Discuss how AI is simultaneously replacing certain jobs and creating new ones, with specific examples of new AI-related careers.
 - c) Evaluate whether AI's overall impact on employment should be viewed as mostly positive, mostly negative, or a mixture of both, and justify your answer.
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Topic 7.2, 7.2.1 & 7.2.2: Data and AI

1. Elaborate on the importance of data in Artificial Intelligence systems.
 - a) Explain why data is often described as the "fuel" of AI, using the student-and-books analogy from the chapter.
 - b) Differentiate in detail between the "quantity" and "quality" of data, and explain why both are necessary for a well-performing AI system.
 - c) Using the dog-and-cat training analogy, construct your own similar example that illustrates how incomplete training data leads to AI errors.
2. Discuss automatic data collection and its role in modern AI systems.
 - a) Explain what automatic data collection means and provide three examples of devices or applications that use it.
 - b) Analyze the benefits automatic data collection provides to AI systems in terms of speed and

responsiveness.

c) Evaluate potential concerns that might arise from devices collecting data automatically and continuously.

Topic 7.3.1: Natural Language Processing (NLP)

1. Construct a comprehensive explanation of Natural Language Processing (NLP) and how it functions.
 - a) Describe the historical significance of ELIZA, including who created it, when, and how it worked.
 - b) Trace, step-by-step, how an NLP system would process the sentence "The service was terrible and slow," identifying tokenization, structural understanding, sentiment detection, and output stages.
 - c) Discuss two real-world applications of NLP and explain the practical benefit each provides to users.
 2. Evaluate the strengths and limitations of NLP-based chatbots in customer service.
 - a) Explain how NLP allows chatbots to answer common customer questions without human involvement.
 - b) Analyze a scenario where a chatbot might misinterpret a customer's message, and explain why this might happen.
 - c) Propose two improvements that could make an NLP chatbot more accurate and reliable.
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Topic 7.3.2: Robotics in Everyday Life

1. Discuss in detail the role of robotics as an application of Artificial Intelligence.
 - a) Describe the historical significance of Unimate, including the task it performed and why it mattered for industrial automation.
 - b) Explain the three-step loop (sensing, deciding, and acting) that most robots follow, using a specific example such as a robotic arm on an assembly line.
 - c) Evaluate the benefits and possible risks of using robots for tasks such as elderly care or hospital surgery assistance.
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Topic 7.3.3: Speech Technology and Voice Control

1. Analyze the role of speech technology in modern devices and daily life.
 - a) Explain how speech technology converts spoken words into text and generates a spoken response.
 - b) Discuss three different real-world contexts (e.g., smart homes, healthcare, customer

service) where speech technology is applied, explaining the specific benefit in each case.

c) Evaluate a scenario where a voice assistant repeatedly misunderstands a particular user's accent, and discuss what this reveals about the system's training data and design.

Topic 7.3.4: Recommendation Systems

1. Construct a detailed explanation of how recommendation systems work and their impact on user experience.
 - a) Explain the historical significance of Amazon's 1998 Item-to-Item Collaborative Filtering system.
 - b) Trace the full recommendation pipeline — from a user's action, through pattern matching and filtering, to the final top 3 predictions — using a specific example of your choice (e.g., a music app or shopping site).
 - c) Discuss the ethical concerns of recommendation systems, particularly regarding privacy and the risk of showing users repetitive or narrow content.
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Topic 7.3.5: Autonomous Vehicles (Self-Driving Vehicles)

1. Discuss in detail how autonomous vehicles use Artificial Intelligence to operate safely without a human driver.
 - a) Construct a full sensor map of a self-driving car, describing what each of the following detects: cameras, radar, LiDAR/proximity sensors, and GPS.
 - b) Analyze the key benefits of autonomous vehicles, including their impact on road safety and accessibility for the elderly or disabled.
 - c) Evaluate the major challenges autonomous vehicles face, including testing requirements, job displacement concerns, and questions of accident responsibility.
 - d) Design a decision-making scenario: describe a hard case (e.g., a pedestrian partially hidden behind a parked vehicle in poor weather) and explain which sensors the car should prioritize and why.
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Topic 7.4 & 7.4.1: Benefits, Challenges, and Limitations of AI Applications

1. Evaluate the overall benefits and challenges of AI applications across different sectors.
 - a) Explain how AI improves efficiency in offices, education, and business planning, with specific examples from the chapter.
 - b) Analyze at least three limitations or challenges of AI applications, including the risks of poor-quality data, job displacement, and unfair decision-making.
 - c) Justify why the chapter emphasizes that AI mistakes are traceable to data, code, or design choices rather than being random or unexplainable.

Topic 7.5, 7.5.1, 7.5.2, 7.5.3 & 7.5.4: Ethics, Fairness, and Social Impact of AI

1. Discuss in detail the concept of bias in Artificial Intelligence systems.
 - a) Define bias in AI and explain, using the automatic soap dispenser case study, how unrepresentative training data leads to discriminatory outcomes.
 - b) Construct a full bias audit for a hypothetical AI-powered building security system trained on 90% male photos, following the steps: stating purpose, checking training data, checking testing methods, predicting the affected group, and proposing a fix.
 - c) Evaluate why the chapter frames identifying bias as "thinking like a responsible tech designer" rather than as a purely negative criticism.
2. Analyze the relationship between fairness and transparency in AI decision-making systems.
 - a) Differentiate clearly between fairness and transparency, providing one example of each from the chapter.
 - b) Discuss why fairness and transparency are especially critical in high-stakes systems such as school admissions, job interviews, and legal cases.
 - c) Design a set of three rules that an AI-powered school admissions system could follow to ensure both fairness and transparency.
3. Evaluate the ethical challenges of privacy, data security, and surveillance in AI-powered systems.
 - a) Differentiate between privacy, data security, and surveillance, using clear definitions and examples.
 - b) Discuss the benefits and risks of public surveillance cameras equipped with AI facial recognition, considering both safety and privacy perspectives.
 - c) Propose two safeguards that could help balance public safety with individual privacy in AI-powered surveillance systems.
4. Discuss the concept of AI and social equity, and its importance for fair access to technology.
 - a) Explain what social equity means in the context of AI, according to the chapter.
 - b) Analyze why AI technologies sometimes become available only to wealthier or more educated groups first, and what gap this can create.
 - c) Propose two ways that AI-powered tools (e.g., medical diagnosis apps) could be made more accessible to people in remote or underserved areas.

Topic 7.6 & 7.6.1: Responsibility of AI Designers and Following Ethics in AI

1. Analyze the responsibilities of AI designers in building fair, safe, and ethical AI systems.
 - a) Explain why the chapter states that "AI is only as good as the people who build it."
 - b) Discuss why building trustworthy AI is described as a moral duty rather than purely a technical task.

- c) Design a complete ethical framework for an AI-powered school admissions system, including at least one rule each for avoiding bias, ensuring transparency, and maintaining accountability.
- d) Evaluate a real or hypothetical case study where an AI system caused a social injustice (e.g., biased facial recognition or unfair hiring decisions), and discuss what the responsible designers could have done differently.